

PIA 408 Course Outline

PIA 408 – Monte Carlo Simulation for Project Appraisal

A distribution instead of a point, computed the same way twice.

School	04 – Public Investment
Hours	30 (indicative band 27-33)
Version	1.0 (2026-09-05)
Model	AICTEM PIAS, D:/EDIP/models/PIASModelling
Reference country	Utopia (UTP), UTD 2024 prices
Prerequisites	PIA 407, whose elasticities this course uses to explain its own results, and PIA 404 for the ENPV being simulated.
Audience	Appraisal economists who have to produce or review a probabilistic result, and the officials who have to decide what one means
Pass mark	60 %

What this course is

PIA 407 established how far an assumption would have to move before the recommendation changed. It could not say how likely that movement was. This course can, at a price: it requires a distribution for every variable that matters, and a distribution is a much stronger claim than a range.

The register holds 11 simulations over 4 variables, with 44 distributions and a correlation structure applied through a Gaussian copula. The engine records the iteration count, the seed, the engine version and the parameter set version on every run, so a result can be reproduced exactly.

Its central finding is one that most practitioners never confront. On the reference project the deterministic ENPV is 218.5 bn and the simulated mean is 28.9 bn – a fall of 86.8%. That is not simulation noise. The input distributions are not centred on the base case: the capital distribution has a mean 12.3% above it and the benefit distribution a mean 7.3% below it. Applying those shifts to the PIA 407 elasticities predicts 32.3 bn, against a simulated 28.9 bn. The gap between a deterministic appraisal and a probabilistic one is arithmetic, and it is reconcilable.

What a graduate can do

Specify a distribution for an appraisal variable and say what evidence would justify it; run a simulation and record everything needed to reproduce it; read a distribution of results – mean, median, percentiles, probability of a negative result, value at risk and conditional expected loss; explain why a simulated

mean differs from a deterministic point estimate and reconcile the two arithmetically; apply and interpret a correlation structure through a copula; state what an implicitly zero correlation asserts; read an iteration count off a reported probability; distinguish a probability of zero from a probability below the resolution of the run; and write a probabilistic result section that does not overclaim.

What this course does not claim

This course does not supply distributions. Every distribution in the register records its source as “SAMPLE_DATA reference-class distribution”, which is a placeholder. A distribution is a claim about how a variable behaves and it needs evidence – a reference class, an engineering assessment, or a hazard model. The course teaches what evidence would justify each shape and says plainly that none of these has it.

It does not model tail dependence. The correlation structure is a Gaussian copula, which reproduces average co-movement and understates the tendency of variables to move together in extreme outcomes. Where the tail is the question – and in a risk analysis it usually is – a Gaussian copula is a step, not a destination.

It does not validate the model being simulated. A simulation propagates uncertainty through a model; it says nothing about whether the model is right. A precise distribution around a wrong answer is still a wrong answer.

It does not make a decision. A probability of a negative ENPV of 39.8% crosses the warning threshold of 30.0% and not the failure threshold of 50.0%. What follows from that is a governance question, and the thresholds themselves are governed parameters with a recorded source like any other.

Learning outcomes

1. State what a simulation adds to a sensitivity analysis, and what it costs.
2. Specify a distribution for an appraisal variable and say what evidence would justify it.
3. Record everything needed to reproduce a simulation exactly.
4. Read an iteration count off the resolution of a reported probability.
5. Read a distribution of results: mean, median, percentiles, and the probability of a negative result.
6. Distinguish a probability of zero from a probability below the resolution of the run.
7. Explain why a simulated mean differs from a deterministic point estimate.
8. Reconcile the two arithmetically using elasticities.
9. Apply a correlation structure through a copula and say what the copula preserves.
10. State what an unspecified correlation asserts.
11. Interpret value at risk and conditional expected loss.
12. Write a probabilistic result section that does not overclaim.

Module structure

Module	Title	Sessions	Hours	Laboratory	Manual	Assessment
M01	What a simulation adds	1	1.25	–	1	Quiz M01
M02	Specifying a distribution	1	3.25	Lab 01	2	Quiz M02

Module	Title	Sessions	Hours	Laboratory	Manual	Assessment
M03	Reproducibility and the run record	1	1.50	–	3	Quiz M03
M04	Iterations, resolution and convergence	1	3.50	Lab 02	4	Quiz M04
M05	Reading the output distribution	1	1.50	–	5	Quiz M05
M06	Zero is not zero	1	3.50	Lab 03	6	Quiz M06
M07	Why the mean is not the base case	1	1.50	–	7	Quiz M07
M08	Reconciling the two	1	1.50	–	8	Quiz M08
M09	Correlation and the copula	1	3.50	Lab 04	9	Quiz M09
M10	The pairs nobody specified	1	1.50	–	10	Quiz M10
M11	Value at risk and expected loss	1	3.50	Lab 05	11	Quiz M11
M12	Writing the probabilistic section	1	1.50	–	12	Quiz M12

Laboratories

Lab	Title	Hours	Deliverable
1	Specify a distribution set, and defend every claim in it	2.0	An annotated distribution set
2	Read the iteration count off the probability, then test convergence	2.0	A resolution and convergence report
3	Bound the zeros, and write the sentences that report them	2.0	A bounded report of the extreme probabilities

Lab	Title	Hours	Deliverable
4	Vary the correlation, and make the implicit zeros explicit	2.0	An explicit dependence structure
5	Size a contingency from the tail, and say what it rests on	2.0	A sized contingency with its assumptions

Assessment

Component	Weight
Knowledge quizzes	15 %
Laboratory exercises	30 %
Applied assignments	25 %
Capstone / final project	30 %

Applied assignments

ID	Title	Weight	Deliverable
A1	The distribution set, defended	25	Marked on separating the three claims and on testing the upper bound
A2	Resolution, reproduction and bounds	25	Marked on the fingerprint check and on never reporting a zero
A3	The reconciliation	25	Marked on the arithmetic and on the explanation of the residual
A4	Dependence and the tail	25	Marked on the sensitivity sweep and on naming all three biases

Final project

A complete probabilistic result section for one project: the input distributions with the evidence each would require and the source actually recorded; the correlation structure, with the pairs left implicitly at zero named; the run record – iterations, seed, engine version, parameter set version – sufficient to reproduce the result exactly; the output distribution with mean, median, percentiles, probability of a negative result, value at risk and conditional expected loss; a reconciliation of the simulated mean against the deterministic ENPV using the PIA 407 elasticities; and a closing paragraph that states what the distribution does and does not establish.

Folder contents

Every course in the Academy carries the same section 5 folder. See `README.md` for what each item holds and `COURSE_BUILD_REPORT.md` for what was measured.